

Mohit Mendiratta

Research Scientist

mohitmendiratta2014@gmail.com +491638693872 Saarbrücken, Germany

https://www.linkedin.com/in/mohit-mendiratta-804604105/ mohitm1994.github.io

https://scholar.google.com/citations?user=Y3Bc8isAAAAJ&hl=en

PROFILE

Research scientist specialising in 3D human modelling, generative AI and multimodal vision-language systems. Most recently, Head of AI Research at QOVES, where I led a team of researchers and interns that built a complete face-editing system in seven months, spanning precise geometric edits, appearance editing, latent control and outpainting. Doctoral research at the Max Planck Institute for Informatics (Dr.-Ing., defence pending), with publications at SIGGRAPH Asia, ECCV, CVPR and 3DV. Experienced in real-time Gaussian head avatars, morphable face models and diffusion-based editing, with a track record of taking research into production.

EDUCATION

09/2021 – 03/2026
Saarbrücken, Germany

Doctor of Engineering (Dr.-Ing.), Computer Science
Max-Planck-Institut für Informatik
Computer Vision and Graphics, focusing on 3D human and facial modelling, generative AI, and multimodal reasoning. Advisor Prof. Dr Christian Theobalt.

10/2018 – 08/2021
Saarbrücken, Germany

Master of Science in Visual Computing
Universität des Saarlandes

08/2013 – 06/2017
Bhubaneswar, India

Bachelor of Technology in Electrical and Electronics Engineering
Kalinga Institute of Industrial Technology

PROFESSIONAL EXPERIENCE

04/2026 – 11/2026
Saarbrücken, Germany

Head of AI Research
Qoves
Led AI research for QOVES's facial analysis and image editing products.

- Built a complete face-editing system in 7 months, spanning precise geometric edits, appearance editing, latent control and outpainting.
- Led a team of researchers and interns.

- Worked with the psychology team to study facial biases in LLMs.

09/2021 – 03/2026
Saarbrücken, Germany

Doctoral Researcher
Max-Planck-Institut für Informatik

- Worked on real-time 3D head avatars using generative models and Gaussian splatting. Built systems for text-based editing of 3D head avatars, audio-driven facial animation, and avatar generation controlled by identity and expression.
- Worked on diffusion models and vision-language models for understanding and editing 3D human faces. Designed and trained models on large datasets and built real-time pipelines.

These works led to publications at SIGGRAPH, SIGGRAPH Asia, CVPR, and 3DV.

08/2019 – 08/2021
Saarbrücken, Germany

Research Assistant
Max-Planck-Institut für Informatik
Developed an egocentric-driven videoconferencing system for XR/AR devices, enabling hands-free video communication on smart glasses and mixed-reality headsets. The system leverages a low-cost wearable egocentric camera module designed for seamless integration into smart eyewear, providing a natural first-person communication experience. This work resulted in a paper published in *ACM Transactions on Graphics (TOG)* and presented at *SIGGRAPH Asia 2020*.

01/2019 – 10/2019
Saarbrücken, Germany

Research Assistant
Fraunhofer IZFP
Developed a no-reference image quality assessment method and a multi-functional ROI analysis tool, along with algorithms for generating large-scale artificial images using heat equation–based modelling.

07/2017 – 08/2018
Bangalore, India

Associate Software Engineer
Zentron Labs Private Limited
Developed image processing algorithms and vision system design for *Fruitron* — India's first automated optical sorting machine using visible, infrared, and ultraviolet imaging with machine learning for high-speed colour, size, weight, and defect sorting.

01/2017 – 06/2017
Bangalore, India

Intern
Zentron Labs Private Limited
Designed and implemented the fruit tracking, ejection, and vision calibration system for *Fruitron*, India's first automated optical sorting machine.

SKILLS

Programming:
Python, C++, CUDA, Bash

Deep Learning:
PyTorch, Lightning, TensorRT

Computer Vision:
OpenCV, COLMAP, Open3D

3D Graphics:
Blender, PyTorch3D, OpenGL, Gaussian Splatting

Vision & ML:
Diffusion Models, NeRF, 3DMM, Face Reconstruction, Multiview Geometry

Systems:
Linux, Docker, Slurm, Multi-GPU Training

AWARDS

03/2020

Saarland Scholarship for International Students
Germany Scholarship (Deutschlandstipendium)
For outstanding academic performance.

PUBLICATIONS

EmoteGPT: 3D Human Facial Expressions from Natural Language Descriptions
European Conference on Computer Vision - ECCV 2026
Haoran Wang, **Mohit Mendiratta**, Christian Theobalt, Adam Kortylewski
Presents a vision-language model that enables semantic reasoning and dialogue about 3D human facial expressions.

Audio-Driven Universal Gaussian Head Avatars
ACM SIGGRAPH Asia Conference Proceedings (SIGGRAPH Asia 2025)
Kartik Teotia, Helge Rhodin, **Mohit Mendiratta**, H. Kim, Marc Habermann, Christian Theobalt
A real-time Gaussian splatting framework for generating high-fidelity 3D head avatars driven by speech audio.

GRMM: Real-Time High-Fidelity Gaussian Morphable Head Model with Learned Residuals
International Conference on 3D Vision 2026 (3DV)
Mohit Mendiratta, Mayur Deshmukh, Kartik Teotia, Vladislav Golyanik, Adam Kortylewski, Christian Theobalt
Introduces a Gaussian morphable head model with learned residuals for expressive, real-time photorealistic face rendering.

Zero-Shot Video Semantic Segmentation based on Pre-Trained Diffusion Models
The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2025
Qian Wang, Abdelrahman Eldesokey, **Mohit Mendiratta**, Fangneng Zhan, Adam Kortylewski, Christian Theobalt, Peter Wonka
Proposes a zero-shot video segmentation method leveraging pre-trained diffusion models without task-specific training.

TEDRA: Text-based Editing of Dynamic and Photoreal Actors
International Conference on 3D Vision 2024
Basavaraj Sunagad, Heming Zhu*, **Mohit Mendiratta***, Adam Kortylewski, Christian Theobalt, Marc Habermann (* equal contribution)
A text-driven framework for editing dynamic 3D human performances using diffusion and multimodal guidance.

AvatarStudio: Text-driven Editing of 3D Dynamic Human Head Avatars
ACM Transactions on Graphics (SIGGRAPH Asia 2023)
Mohit Mendiratta, Xingang Pan*, Mohamed Elgharib*, Kartik Teotia, Mallikarjun B R, Ayush Tewari, Vladislav Golyanik, Adam Kortylewski and Christian Theobalt (* equal contribution)
Introduces a system for photorealistic, text-based editing of dynamic 3D head avatars using generative models.

Egocentric Videoconferencing
ACM Transactions on Graphics (SIGGRAPH Asia 2020)
M. Elgharib*, **M. Mendiratta***, J. Thies, M. Nießner, H-P. Seidel, A. Tewari, V. Golyanik and C. Theobalt (* equal contribution)
Presents an egocentric video conferencing system enabling hands-free video calls using wearable cameras for smart glasses.

Beauty is in the AI of the beholder: MLLMs systematically overrate facial attractiveness
arXiv 2026
Santiago Grandas, Juan Sebastian Cely-Acosta, Mohit Mendiratta, Shafee Hassan, Macken Murphy
A pre-registered study comparing 2,513 human raters with four commercial MLLMs (Claude, Gemini, GPT and Grok), showing that the models track human rank-ordering of facial attractiveness but systematically rate faces higher and within a narrower range.